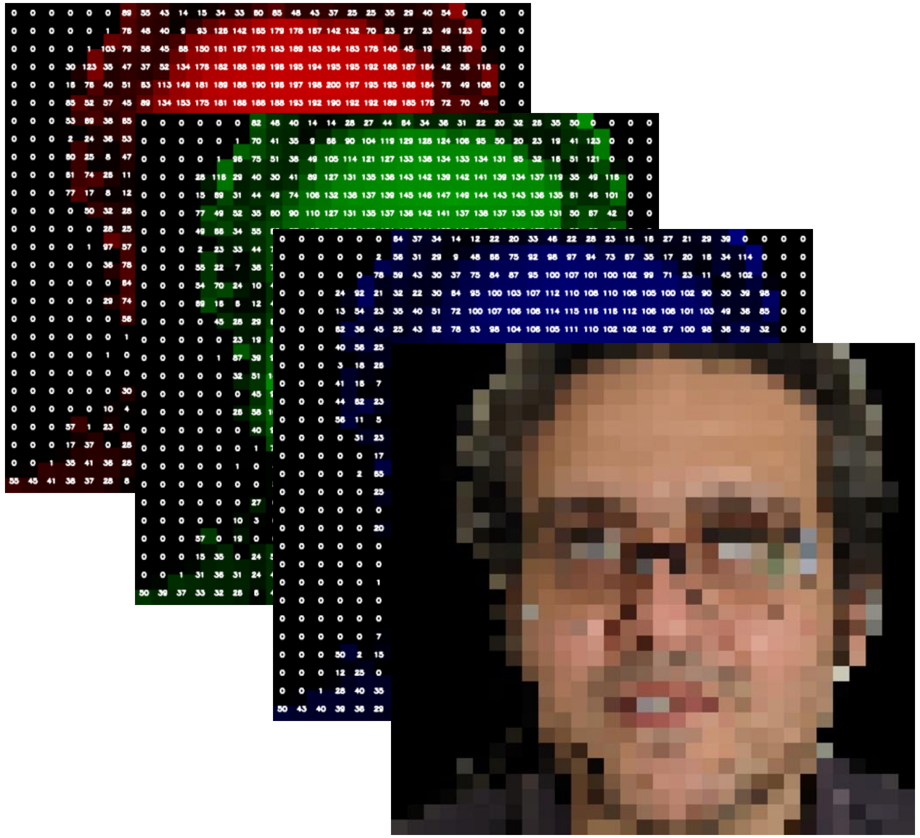
Image Processing - Lecture 1



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Reading:

Gonzalez/Woods – Ch. 1 + 2.1 + 2.4

Szeliski – Section 2.3 (background)

(one of, not all)

Introduction to Image Processing and Image Representations

Slide material acknowledgements: Fisher; Breckon; Ivrissimtzis

Why Process Images?

Vision: a powerful source of information

- information about environment
- **shape, size, colour, position** and **orientation**.

Visual Analysis: infer higher level knowledge

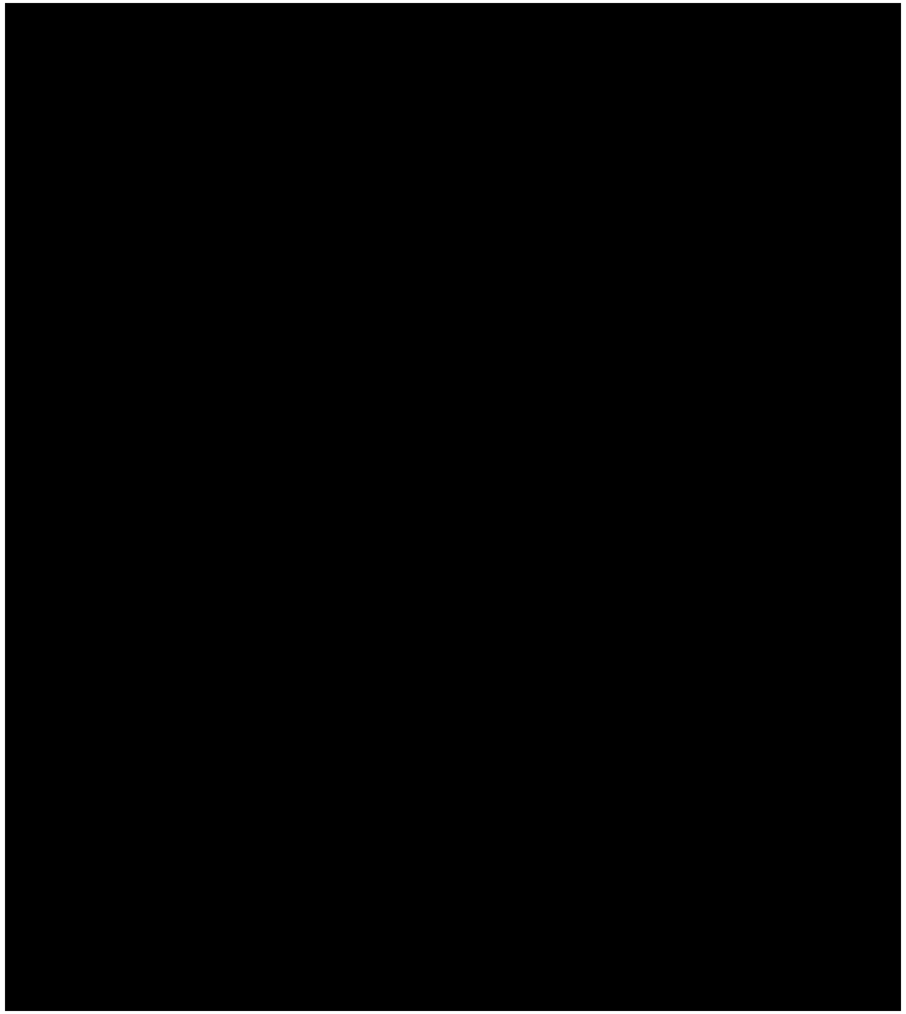
– from environment or visual information presented before us.

Why Process Images?

- Known as the McGurk effect [McGurk & McDonald, 1976]
- Demonstrates that we actually use vision (image sensing) in tasks that are assumed to be non-visual (e.g. listening to speech).

- **Conclusion:** processing visual information is important!

Video: https://youtu.be/XFtWOak4ZKI



Processing Visual Information

- An image contains visual information.
- Image processing is about processing visual information.
- Why?
  - to extract knowledge automatically
  - to enable humans to extract information
  - to make it more appealing visually

What Is an Image ?

Captured graphical representation of a **scene**.

Snap-shot in time.

2D signal.

Recording of visual information.

What Does an Image Contain?

Image encodes:

- edges / lines
- regions
- orientations
- scale
- .....
- lighting
- .....
- noise!

.... using a 2D signal.

Application Themes

Assistive Imaging: **enhancement**, **restoration**, **representation** or **transformation** of visual data to aid in visualisation and interpretation by humans or as pre-processing step for computer vision.

- the focus of **this course**

Computer Vision: automatic interpretation of visual data using computers, without human intervention.

- covered in Level 3 (and 4) Computer Science modules

Where Does It Fit ... as a Science?

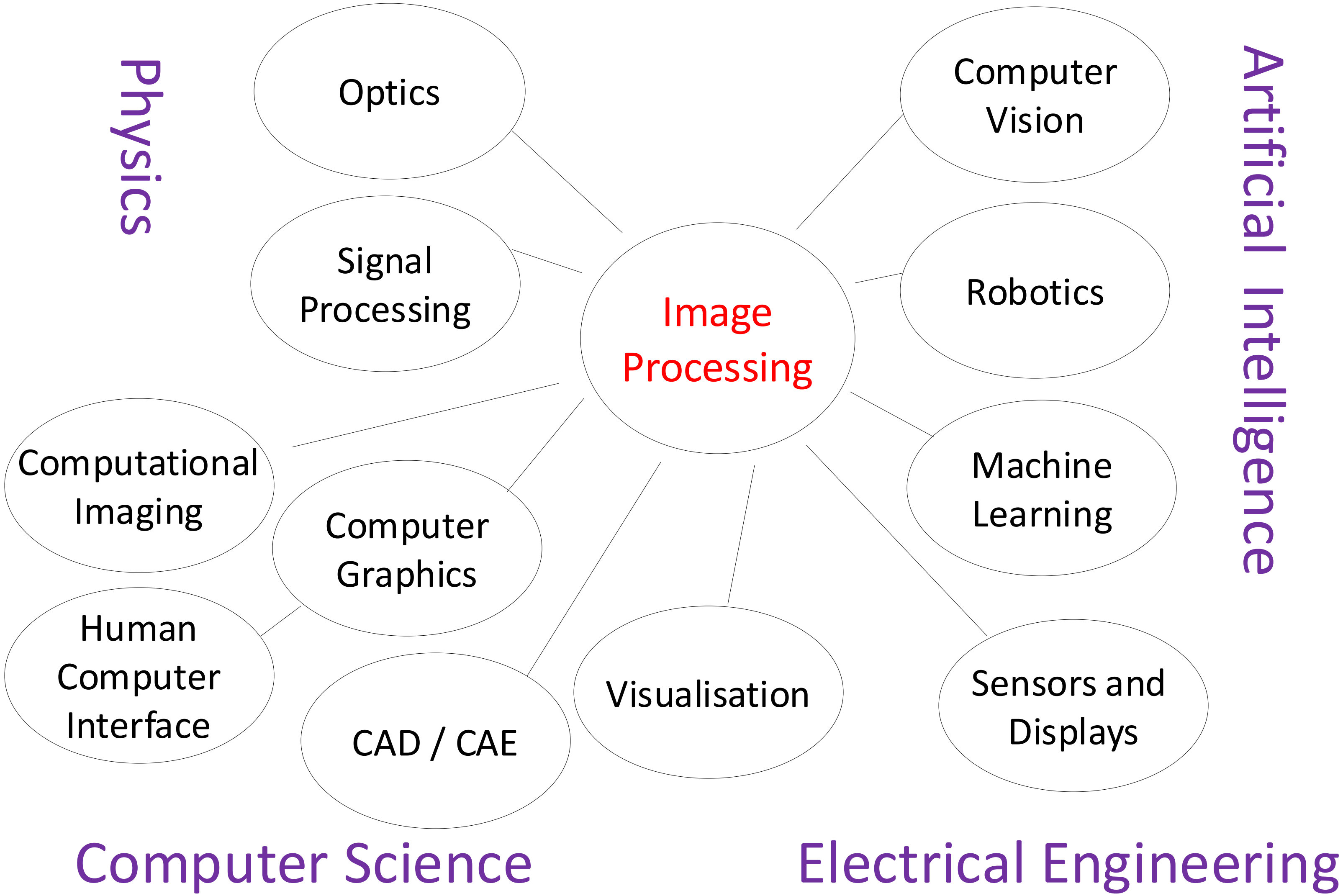


Image Representation

An image is:

- A **multidimensional signal**, commonly containing visual information

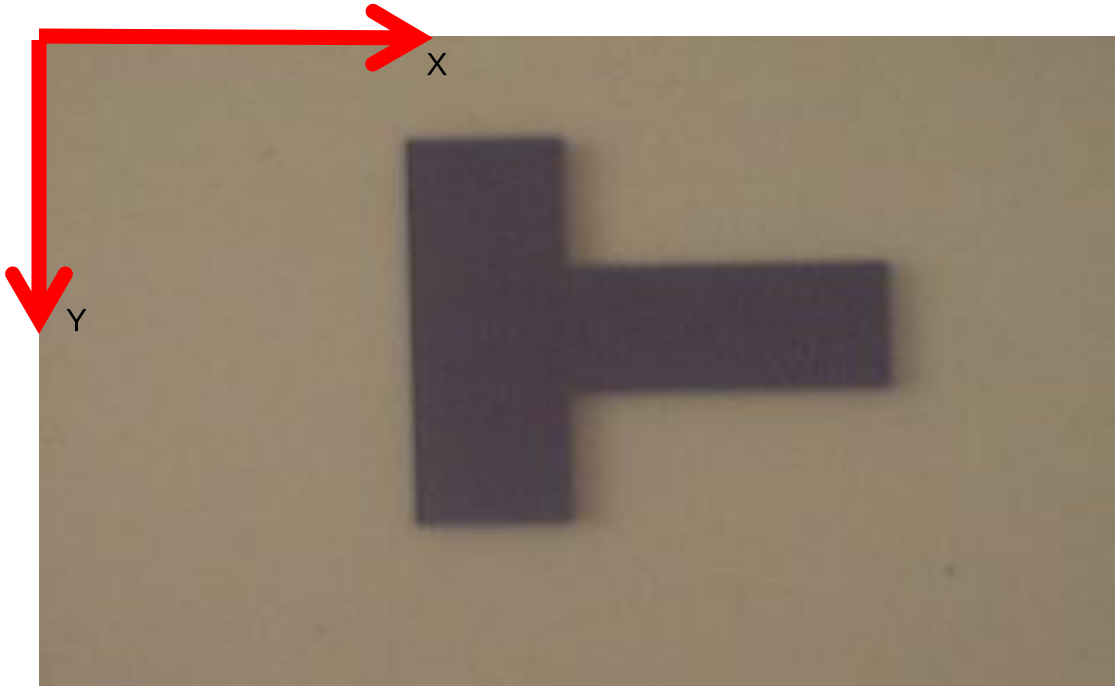
Represented by

- 2D grid of values
  - indexed as (x, y) into the grid – i.e. pixel location

  Pixel = picture element.

(A two-dimensional function, $f(x, y)$, where the value of f at (x, y) is scalar quantity)

Image Spatial Resolution



Spatial resolution:

- $X \times Y$ (horizontal by vertical) dimensions of the image.

- Number of pixels used to cover the visual space captured by the image.
- Relates to the sampling of the image signal.
- Commonly quoted as $X \times Y$ (e.g. 640×480, 800×600, 1024×768, etc.)

Quick Demonstration

- Change Image Resolution:

https://github.com/atapour/ip-python-opencv/blob/main/lower_resolution.py

Image Colour Resolution



- Colour resolution: dimension of the colour space
  - – known as quantisation.
- Represents the number of **possible intensity/colour values** that a pixel may have.

Image Colour Resolution



- Binary image has 2 colours (black or white).
- Grayscale commonly has 256 grey levels.
- Colour commonly has 256 values per colour channel (**R**, **G**, **B**).

➢ Often quoted as number of binary bits:

binary = 1-bit, grayscale = 8-bit, colour = 24-bit.

Quick Demonstration

- Change Colour Resolution:

https://github.com/atapour/ip-python-opencv/blob/main/quanise_colours.py

Temporal Resolution (Video)

- In continuous capture systems (e.g. video), temporal resolution is the **number of images captured in a given time period**.

- Commonly quoted in **frames per second (fps)**
  - each image is referred to as a video frame.

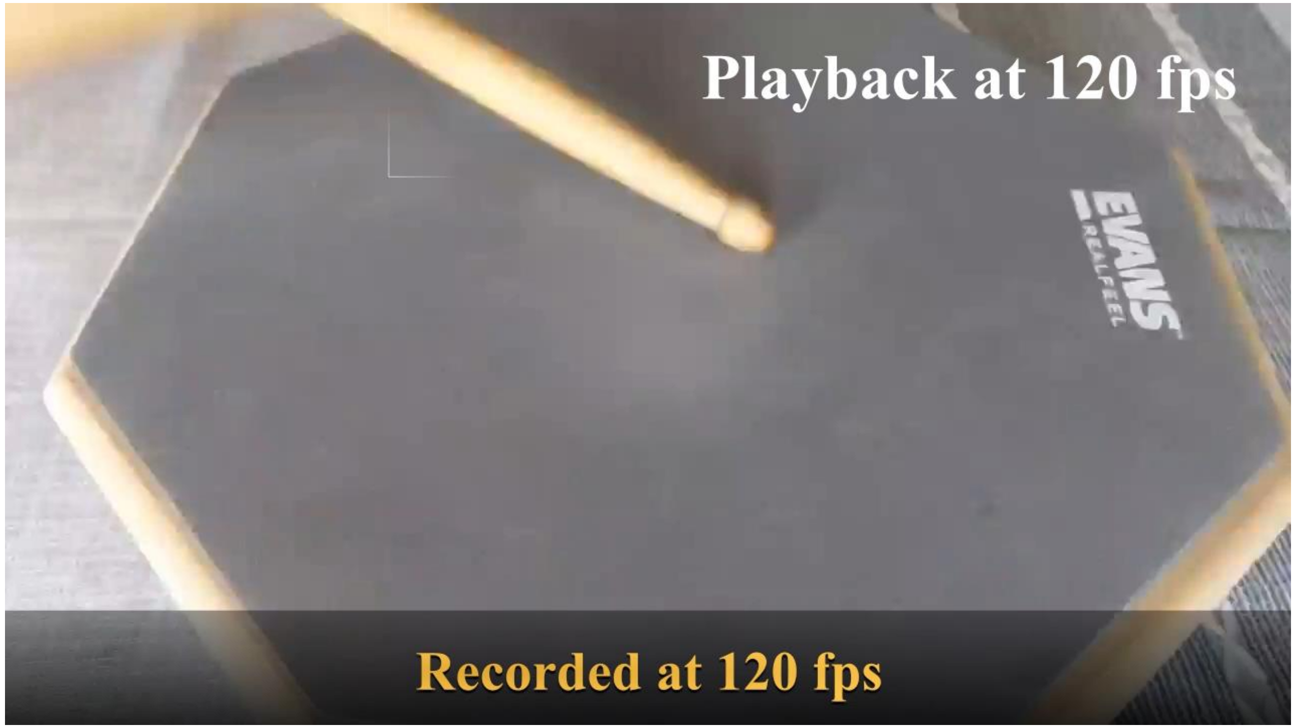
Temporal Resolution (Video)

- TV in UK/Europe: 25 fps
- TV in US/Asia: 30 fps
- Cinema: 24 fps

- 25-30 fps is suitable for most visual surveillance applications
- higher framerates available for specialist scientific/engineering applications, gaming and recently, ... very very common!

High Temporal Sampling

Source: anon. (lost to the myths of time)



Source: recorded using a mobile phone

https://youtu.be/zIi-rV-9I2M

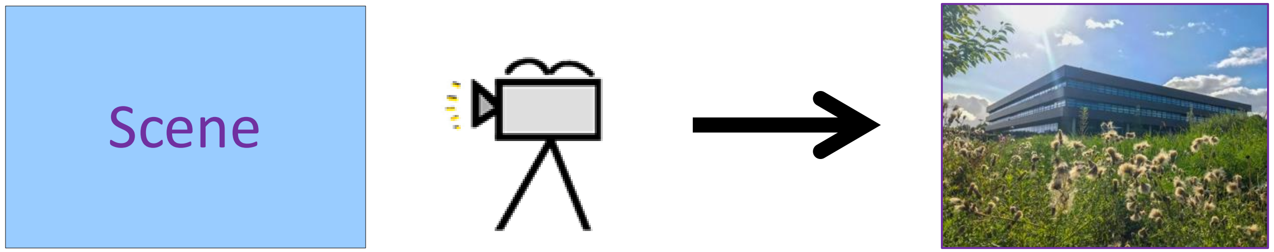
High framerate cameras are now readily available.

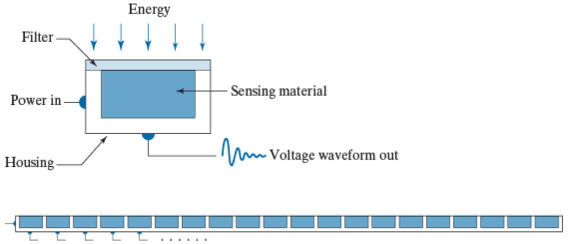
Quick Demonstration

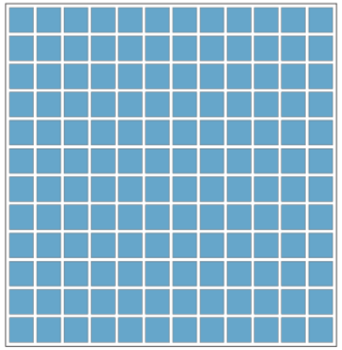
- Change Framerate:

https://github.com/atapour/ip-python-opencv/blob/main/change_framerate.py

Representational Requirements







Sensor array

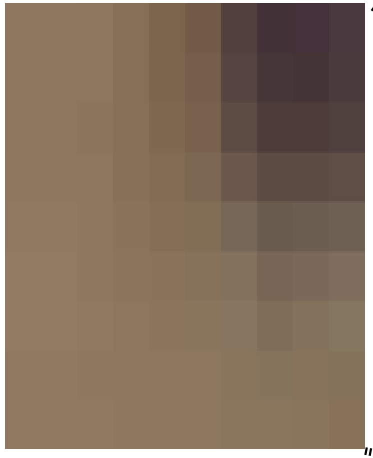
Scenes have to be sampled (**spatial**, **temporal**) and **quantised**.

- analogue to digital conversion.
- Real world signals (light) → discrete form (digital image).

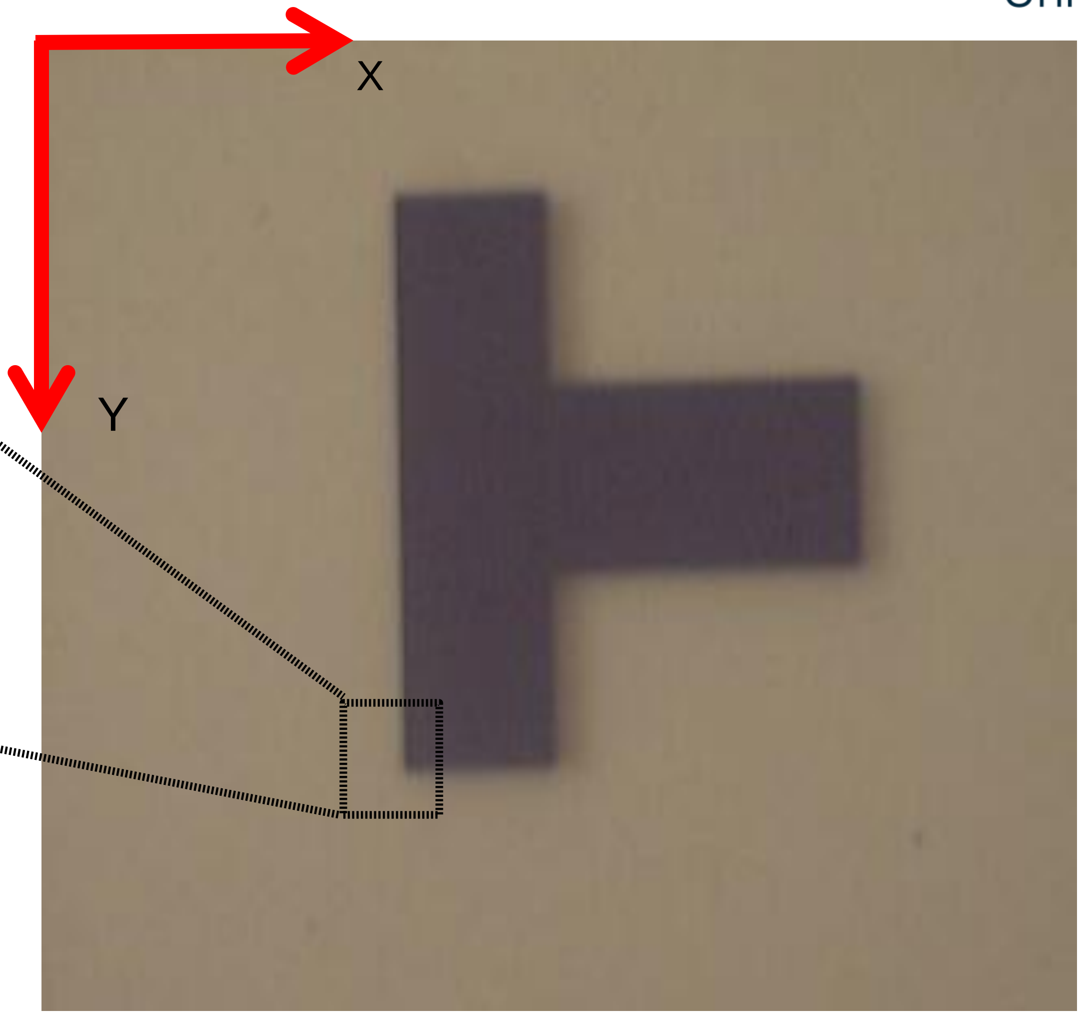
Requirement:

- Sampling must be high enough to preserve useful information in the image.
- Quantisation must avoid aliasing.

Resolution **Is** Limited



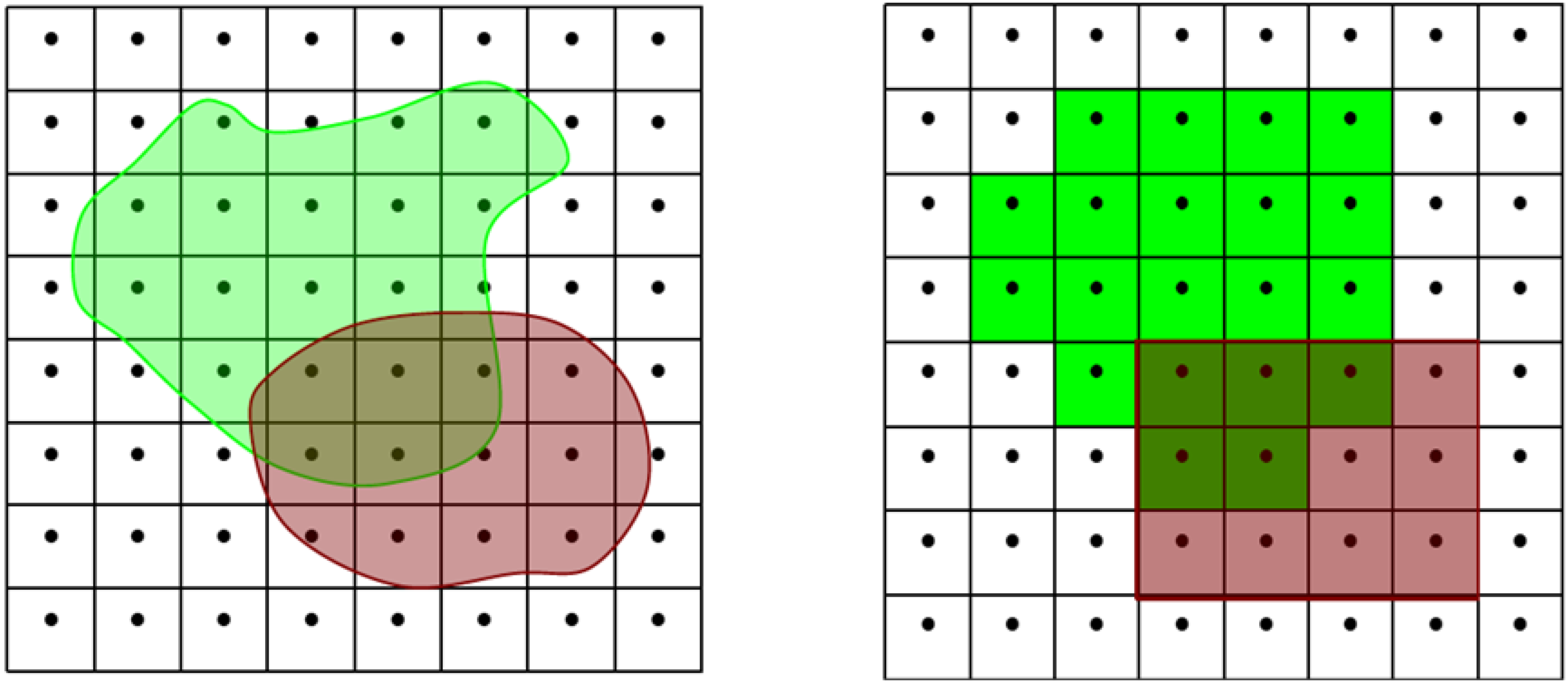
Limitations of
discrete representation
(i.e. camera resolution)



In all 3 modalities - **space**, **colour**, **time** - this causes **aliasing** effects.

Image Aliasing

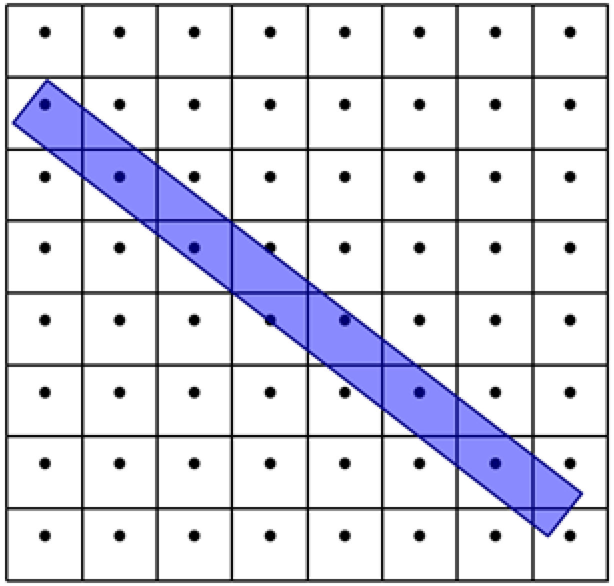
Analogue image signals always see some aliasing in the digitisation process.



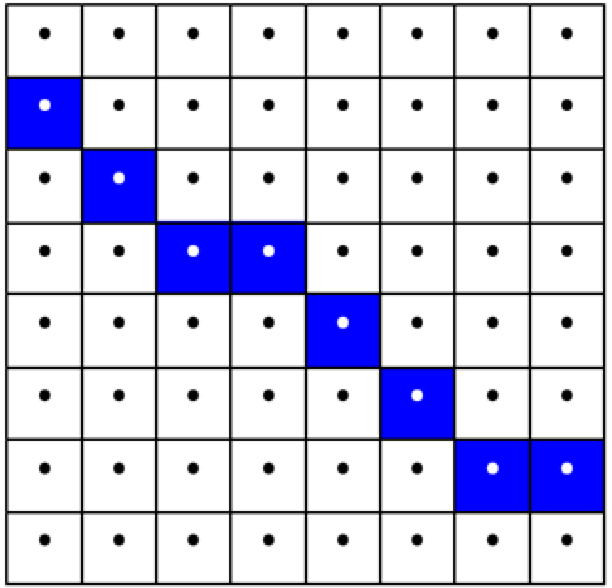
analogue signal (world) ⇒ sampling ⇒ digital signal (image)

Image Aliasing

Aliasing effects both the connectivity and topological image features.



noise *is introduced*



analogue signal (world) ⇒ sampling ⇒ digital signal (image)

Image processing algorithms must be able to cope with problems arising from this form of **sampling noise**.

Varying Spatial Sampling

Original Image
(HxW = 128x128)

Image 2
(HxW = 64x64)

Image 3
(HxW = 32x32)

Image 4
(HxW = 16x16)

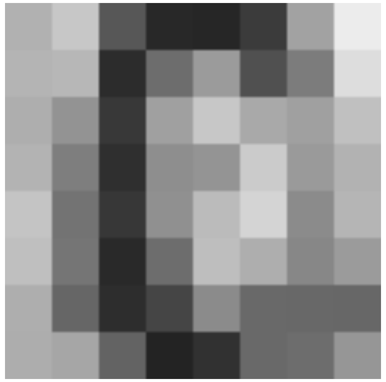


Image 5
(HxW = 8x8)

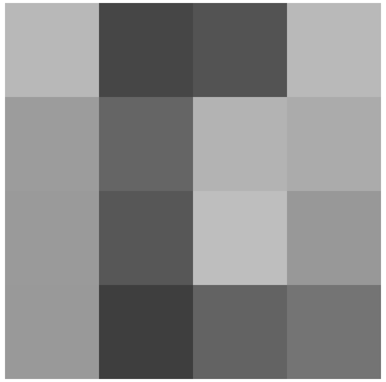


Image 6
(HxW = 4x4)

Less detail means less information to extract from the image.

Also, fewer pixels to process!

Colour (or Intensity) Quantization

At each pixel:

- a voltage reading on image sensor that relates to the amount and wavelength of light received.
- is discretised into a number of bins representing a level of intensity (pixel values).

For example, in an 8-bit grayscale sensor:

- 2^8 bins = 256 possible pixel values = 8-bit integer value
- representing grayscale ranging from

0 (black) to 255 (white)

256 grey-levels

Varying Intensity Quantization

256 grey-levels

128 grey-levels

64 grey-levels

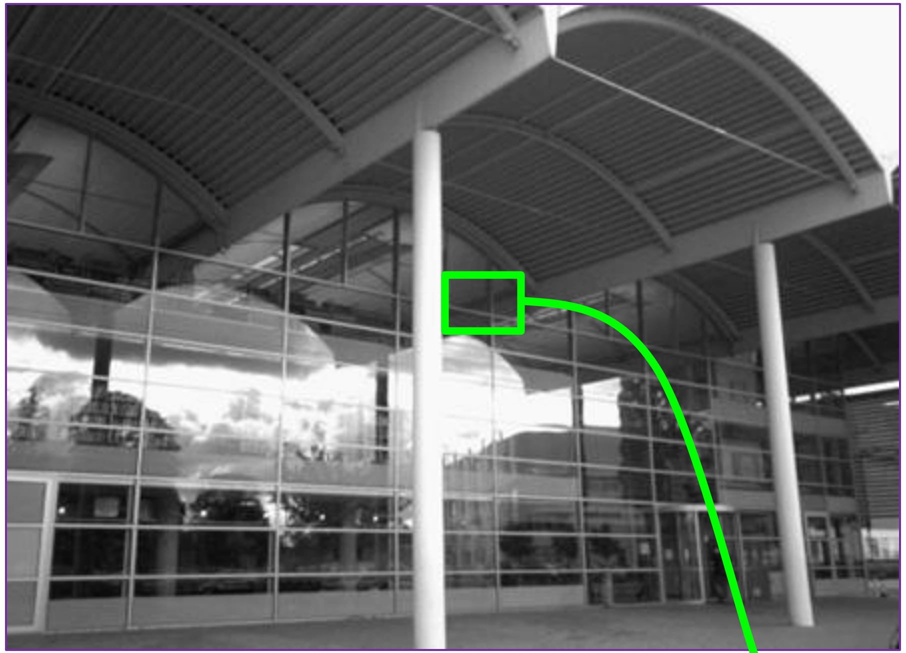
32 grey-levels

8 grey-levels

4 grey-levels

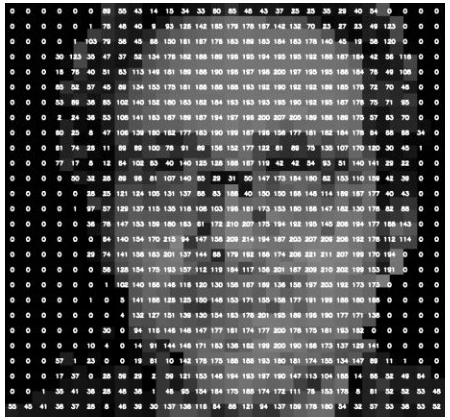
- Human eye can "see" **40 grey levels** and **7-10 million colours**.
- Important to know what we can get away with before **colour aliasing** is notable.

Image Representation



Images are 2D signals.

Signals can be digitised as a **structured set of numbers**.



Images are often represented as **2D arrays** or **matrices**.

138	136	67	111	152	82	84	150	106	69	136	131	65	111	150	77	81	152	104
147	125	64	123	148	72	94	155	96	72	146	121	63	124	141	67	95	151	86
79	88	153	104	70	139	130	65	115	150	76	85	152	101	65	137	127	60	114
66	108	154	81	81	150	108	65	136	135	63	109	152	77	78	150	104	60	134
88	79	149	112	67	131	135	66	106	152	80	77	150	105	62	132	131	62	105
91	75	149	116	63	127	140	68	101	154	86	75	148	113	62	128	137	63	99
149	117	63	127	144	68	99	155	89	74	147	115	62	124	138	67	97	150	86
149	121	61	124	144	69	96	154	91	72	145	118	62	120	142	71	93	149	90
140	129	64	114	148	76	86	154	101	66	139	126	62	115	145	71	89	151	94
139	133	64	114	150	75	85	152	101	65	138	126	63	114	145	72	86	151	96

e.g. `image[x][y] = 150`

Addressing Image Pixels

- Image represented by $X \times Y$, column $\times$ row, matrix.

Value at each (x, y) location in the matrix represents the intensity of the image at that pixel.

- grayscale = integer
- colour = (r, g, b) triplet of integers

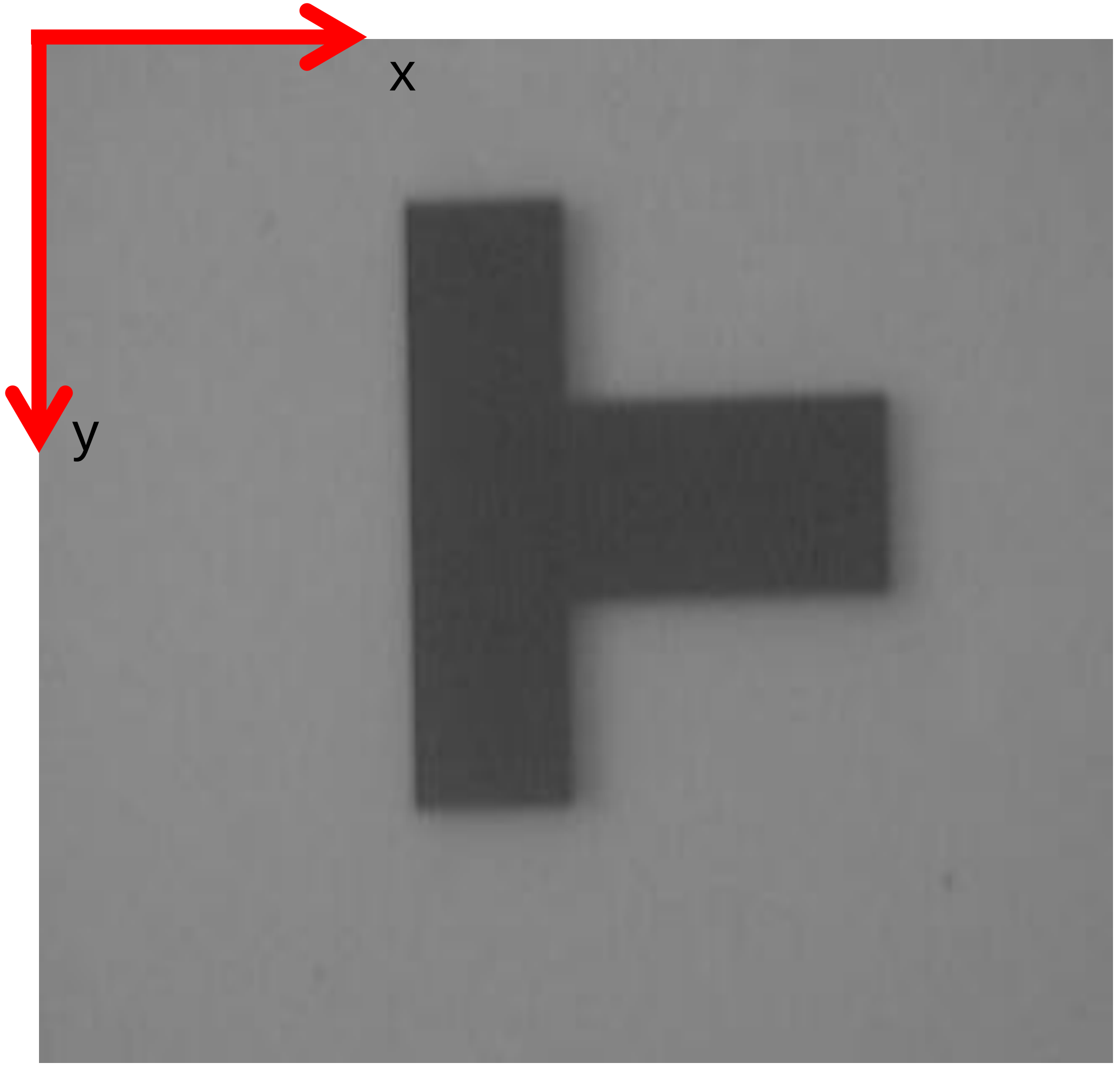
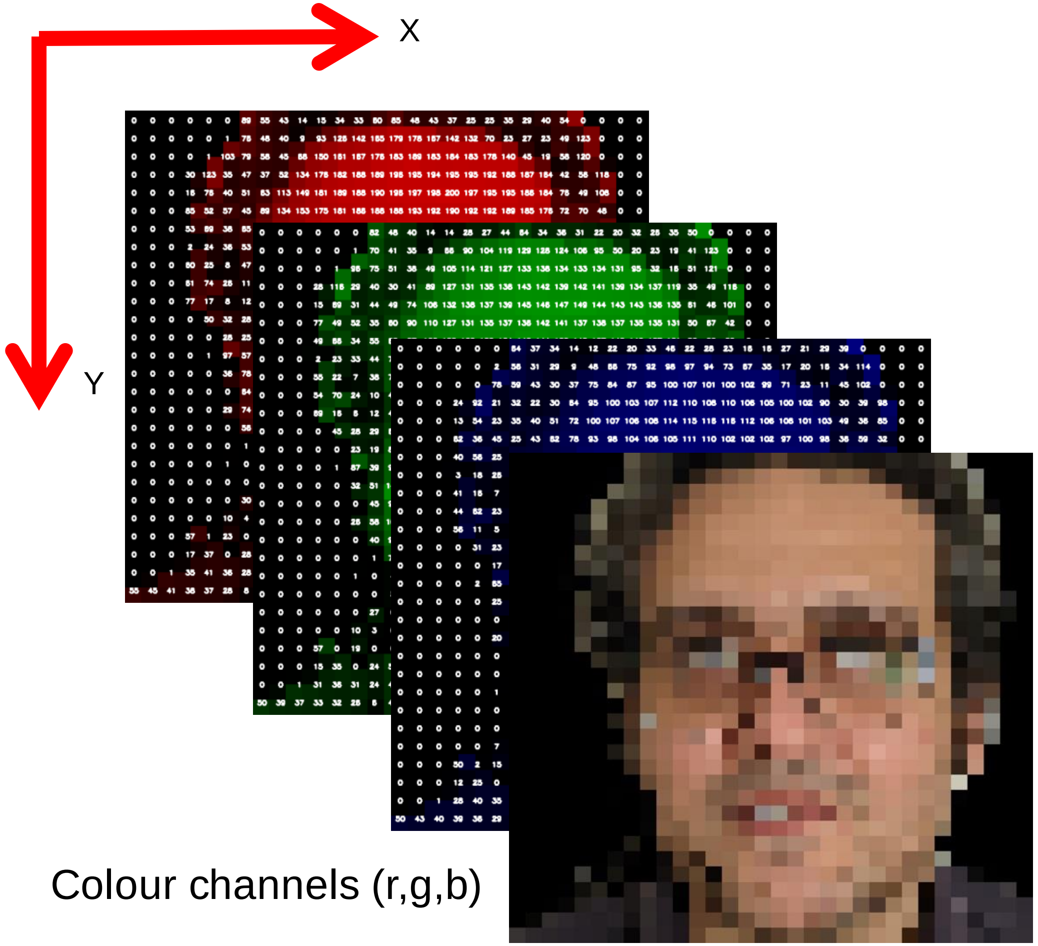


Image Colour Channels



Colour channels (r,g,b)

RGB Image Representation:

- Red, Green and Blue
- light intensity for each (x, y) pixel gives an {r,g,b} integer vector.
- Colour image = 3-channel image (x, y, i) for $i = \{0,1,2\}$.

N.B. Greyscale image = 1 channel.

Beware: some image libraries such as *OpenCV* use BGR by default!

R,G,B colour channels

Pixel Co-ordinates

image (x,y)

- **Greyscale images**: image(x,y) = value.
- **Colour images**: image(x,y) = (r,g,b).
- **Pixel location**: (x,y) position in the column by row co-ordinate system.

Positive co-ordinate system:

- origin = top left.

Origin is not universal: OpenCV uses top left and:

- column (c) = x co-ordinate,
- row (r) = y co-ordinate.

So, what do pixel values actually represent in image data?

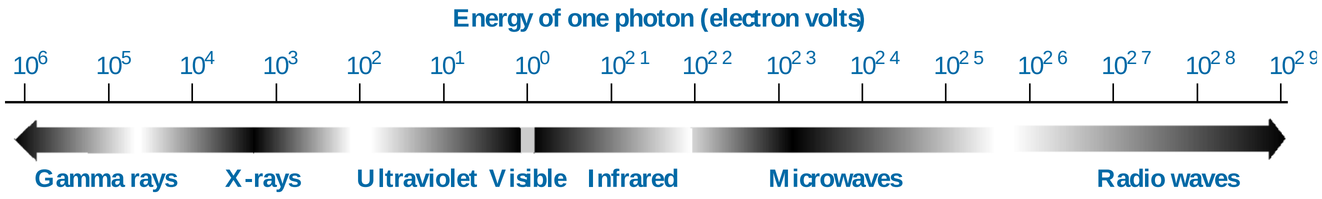


Image Data: Light Intensity / Colour

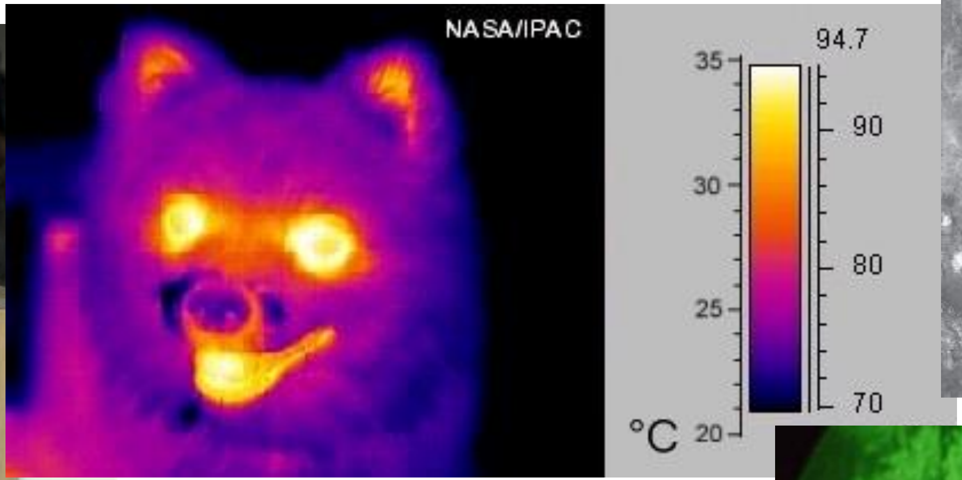
Pixel values encode wavelength (colour) or intensity of light.

Issues:

- Perspective projection to 2D is not easily invertible.
- Quantisation.

Image Data: Infra-Red

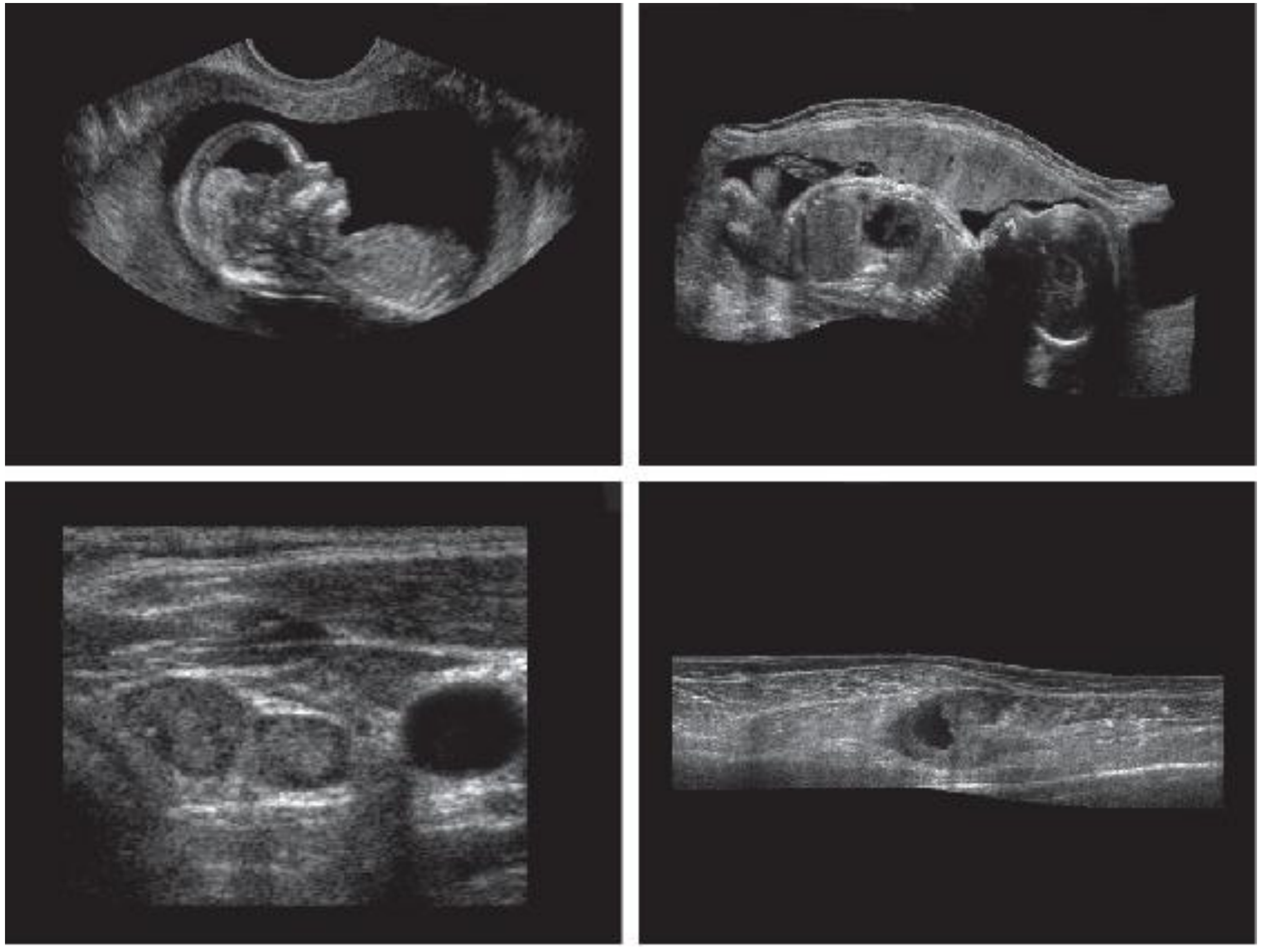
Pixel values encode infra-red electromagnetic intensity.



Issues:

- near and far infra-red (different wavelengths).
- visualisation may require colour mapping.

Image Data: Ultrasonic



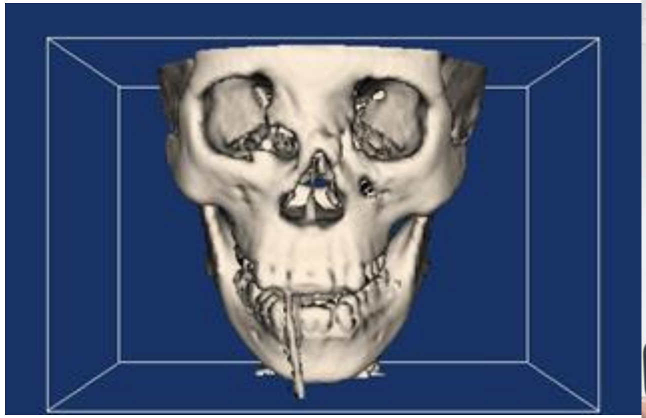
Pixel values encode reflection characteristics of tissues when high-frequency sound waves are sent through the body.

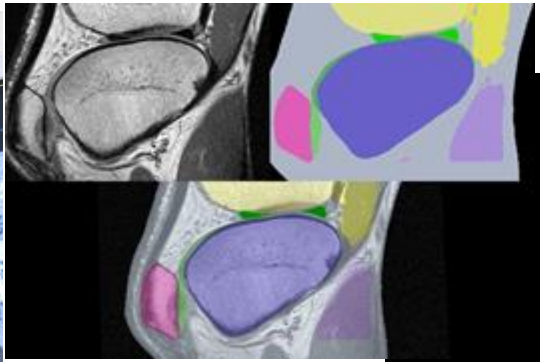
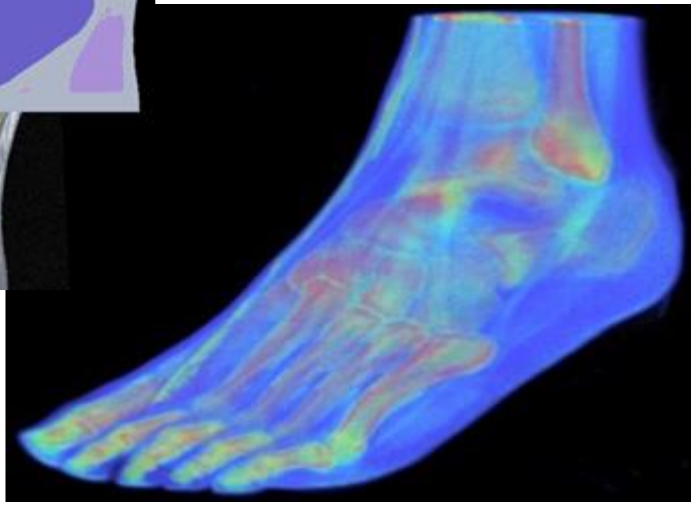
Issues:

- artefacts with regards to location, brightness, shape, and size of objects.
- visualisation may require colour mapping.

Image data: Medical CT / MRI / PET

Pixel values are proportional to the absorption characteristics of tissue in relation to a signal sent through the body.



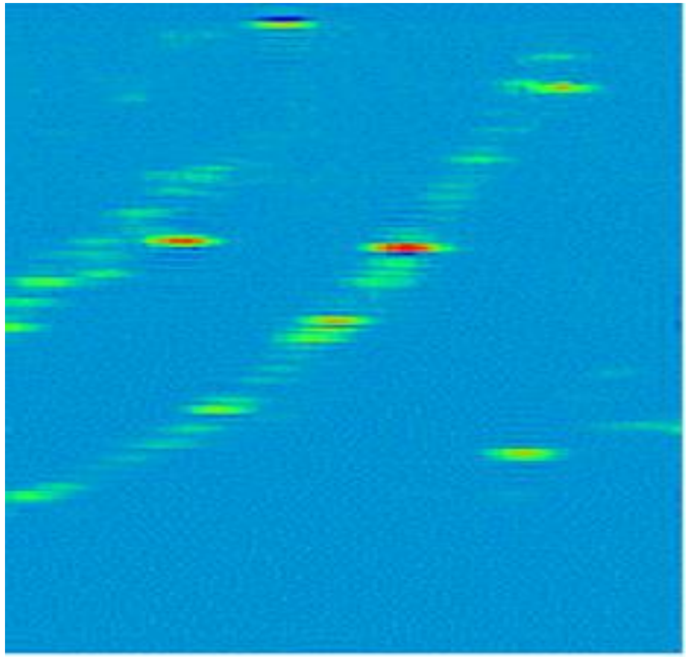



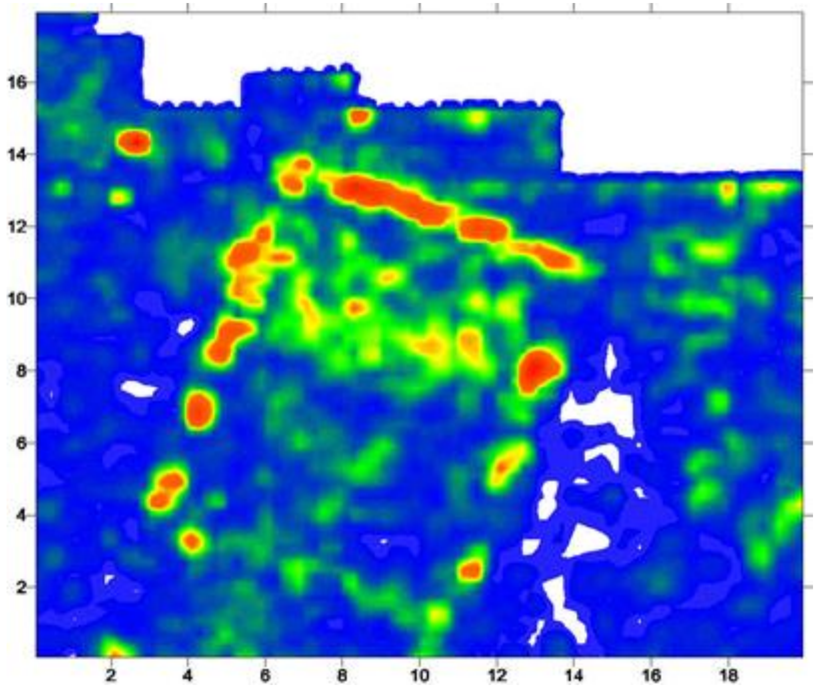
Issues:

- segmentation.
- visualisation of volumetric data (medical imaging).

Image Data: Radar

Pixel values are proportional to target distance from sensor and reflectivity.



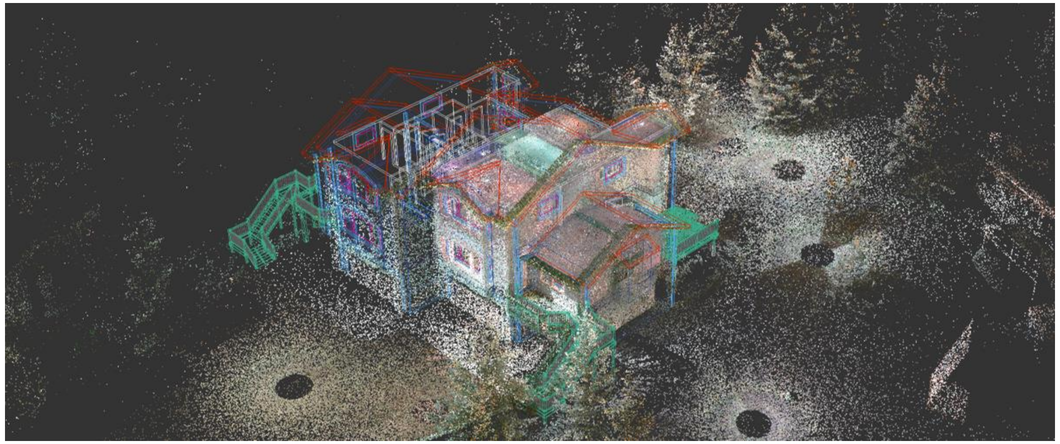


Issues:

- noise and artefacts.
- calibrating values to correspond to distance.

Image Data: Depth / Distance

Pixel values encode distance of object / surface from sensor.



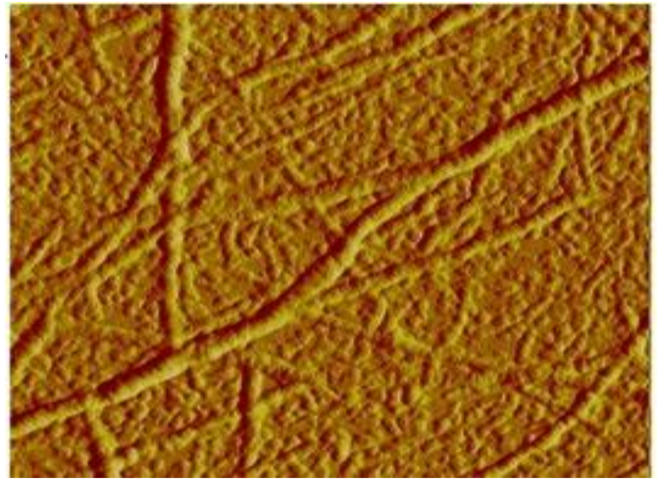
[Atapour et al., 2019]

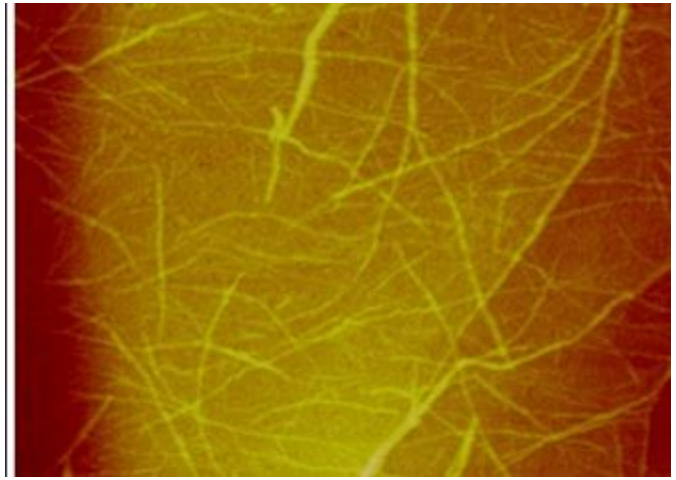
Issues:

- partial view (2½D) of the captured 3D object.

Image Data: Scientific

Pixel values encode measurements from a given sensor.



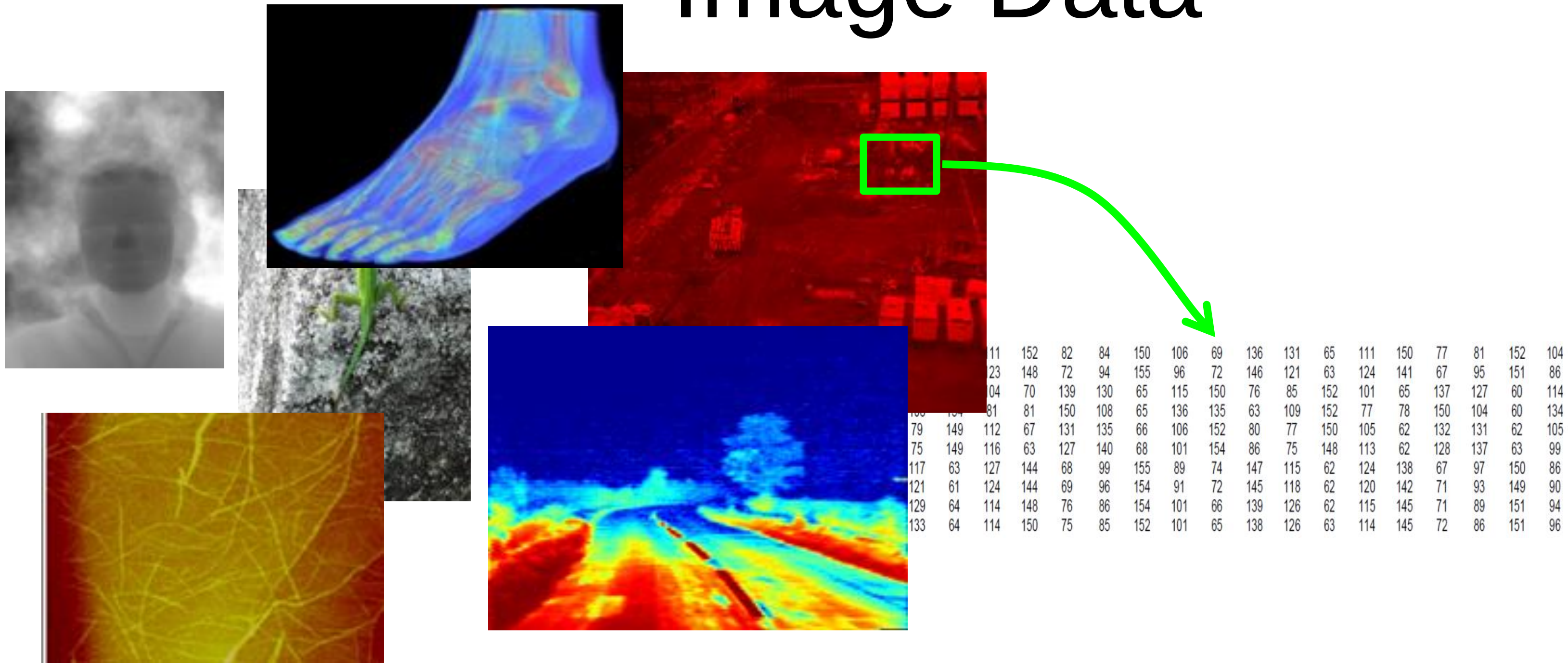


Atomic Force Microscopy images of proteins forming fibres. The colours refer to height.
R. Higham – University of Edinburgh

Issues:

- representation: positive and negative floating point image values
- visualisation: colour mapping / negative value scaling

Image Data



Diverse applications and datatypes, but are all just

- numbers of some form or another
  - same topology: regular grid = image.

So, we can apply **common processing** methodologies to ***all*** of them.

Summary

- **Image Processing Background**

- **Image Sampling / Resolution**

  – **spatial resolution** = spatial sampling *(how many pixels?)*
  – **temporal resolution** = temporal sampling *(how many frames?)*
  – **colour resolution** = colour sampling *(how many intensity/colour levels?)*

- *Differing image types / applicability of image processing*

- Practical sessions are important!
- Know time and location of your practical.

Please watch the video on the module page on Ultra!

Some Housekeeping